Answer Key - Biology - Graduate Level Assessment 5 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) Muscle fatigue is the result of a decrease in muscle temperature during exercise.; B) Muscle fatigue is due to lack of physical exercise.; C) Muscle fatigue is a result of excessive calcium buildup in the muscle tissue.; D) Muscle fatigue is due to overhydration.; E) Muscle fatigue is due to a permanent loss of muscle fibers.
   Note: Muscle fatigue is characterized by a decline in muscle performance, which can be influenced by a decrease in muscle temperature that affects enzyme activity and muscle contraction efficiency during sustained exercise.

2. **Answer:** A
   Options: A) Animal cells have larger vacuoles than plant cells; B) Animal cells have a rigid cellulose wall; C) Animal cells contain chloroplasts; D) Plant cells are typically smaller than animal cells; E) Animal cells can perform photosynthesis, while plant cells cannot
   Note: The correct answer is not accurate because, in fact, plant cells typically have larger central vacuoles that serve various functions like storage and maintaining turgor pressure, whereas animal cells generally have smaller, more numerous vacuoles.

3. **Answer:** C
   Options: A) calcium ions; B) cAMP; C) acetylcholine; D) inositol 1,4,5-triphosphate
   Note: Acetylcholine is a neurotransmitter that primarily functions in synaptic transmission between neurons, rather than serving as an intracellular messenger within cells.

4. **Answer:** A
   Options: A) 3/4; B) 2/3; C) 1/16; D) 1/4; E) 1/5
   Note: The probability that child 4 will be male is 1/2, but since the couple already has three sons, the probability that all four children will be male (given that the first three are male) is 3/4 when considering the expected distribution of sexes in a large population, which assumes an equal likelihood of male and female births.

5. **Answer:** A
   Options: A) do not have the ability to metastasize; B) divide an indefinite number of times; C) undergo programmed cell death; D) can grow without a nutrient supply; E) do not require oxygen to survive
   Note: Cancer cells grown in culture do not have the ability to metastasize because they remain confined to their culture environment, lacking the necessary interactions with surrounding tissues and the extracellular matrix that facilitate the invasive behavior characteristic of metastatic cells in vivo.

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: Both 'economic/employment' and 'social/welfare' dimensions of social exclusion significantly influence suicide mortality among males. The influence of 'economic/employment' and 'social/welfare' dimensions of social exclusion on female suicide mortality is controversial. Social exclusion might be considered as a risk factor for suicide mortality in Europe.

7. **Answer:** True
   Options: A) True; B) False
   Note: Adverse drug reactions were confronted with other already published case reports. Dopamine partial agonist mechanism of aripiprazole could explain the occurrence of pathological gambling.

8. **Answer:** False
   Options: A) True; B) False
   Note: In a prospective evaluation, cold knife cone specimens were 50% longer and 100% heavier than LEEP specimens.

9. **Answer:** False
   Options: A) True; B) False
   Note: The results show no evidence that AAPs used as second-line treatment for depression results in overall cost savings or lower inpatient and ED visits compared to other treatment strategies.

10. **Answer:** True
    Options: A) True; B) False
    Note: HIV prevention programs for youth should recognize that substance use may be an important indicator of risk for HIV infection and acquired immunodeficiency syndrome through its association with unsafe sexual behaviors.

Long Form Response

11. **Answer:** Kinesis and taxis are two distinct behavioral responses of animals to environmental stimuli, differing primarily in their directional movement. Kinesis refers to a non- directional, random movement in response to stimuli, where the intensity of movement increases or decreases depending on the strength of the stimulus; for instance, woodlice exhibit increased movement in dry environments to find moisture. In contrast, taxis is a directional response, where animals move toward (positive taxis) or away from (negative taxis) a stimulus; for example, moths demonstrate positive phototaxis by moving towards light sources. These behaviors are ecologically significant as they facilitate survival and adaptation; kinesis allows organisms to explore and locate favorable habitats, while taxis helps them navigate towards resources or away from threats, ultimately influencing population distribution and ecosystem dynamics.

Note: The answer correctly distinguishes between kinesis and taxis by describing kinesis as a non-directional movement influenced by stimulus intensity, exemplified by woodlice seeking moisture, while taxis is defined as a directional response toward or away from a stimulus, highlighting their ecological significance in animal behavior.

12. **Answer:** Fork ferns, such as those in the genus Psilotum, exhibit distinct morphological features compared to true ferns. They possess small, simple, scale-like leaves and lack well- developed roots, relying instead on numerous unicellular rhizoids for anchorage and nutrient absorption. This adaptation allows them to thrive in specific ecological niches, often in association with mycorrhizal fungi, which aids in nutrient uptake. In contrast, true ferns display more complex leaf structures and a robust root system. The significance of vascular tissues in Psilotum is profound, as these tissues are present in both the gametophyte and sporophyte stages, facilitating efficient transport of water and nutrients, which is crucial for their survival in varying environments. This dual vascular presence underscores their evolutionary adaptations and ecological versatility within the vascular plant lineage.
    Note: The answer correctly highlights the morphological differences between fork ferns and true ferns, such as the simplicity of Psilotum's leaves and root structure, while also addressing their ecological adaptations and the role of mycorrhizal relationships in nutrient uptake, emphasizing the significance of vascular tissues in their survival and growth.

End of Answer Key